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Transferable Benefits of Cycle Hypoventilation Training for Run-Based Performance in Team-Sport Athletes

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Purpose: To determine whether high-intensity training with voluntary hypoventilation at low lung volume (VHL) in cycling could improve running performance in team-sport athletes. **Methods:** Twenty well-fit subjects competing in different team sports completed, over a 3-week period, 6 high-intensity training sessions in cycling (repeated 8-s exercise bouts at 150% of maximal aerobic power) either with VHL or with normal breathing conditions. Before (Pre) and after (Post) training, the subjects performed a repeated-sprint-ability test (RSA) in running (12 × 20-m all-out sprints), a 200-m maximal run, and the Yo-Yo Intermittent Recovery Level 1 test (YYIR1). **Results:** There was no difference between Pre and Post in the mean and best velocities reached in the RSA test, as well as in performance and maximal blood lactate concentration in the 200-m-run trial in both groups. On the other hand, performance was greater in the second part of the RSA test, and the fatigue index of this test was lower (5.18% [1.3%] vs 7.72% [1.6%]; $P < .01$) after the VHL intervention only. Performance was also greater in the YYIR1 in the VHL group (1468 [313] vs 1111 [248] m; $P < .01$), whereas no change occurred in the normal-breathing-condition group. **Conclusion:** This study showed that performing high-intensity cycle training with VHL could improve RSA and possibly endurance performance in running. On the other hand, this kind of approach does not seem to induce transferable benefits for anaerobic performance.